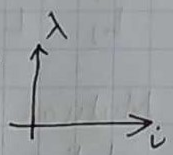
Lecture 22

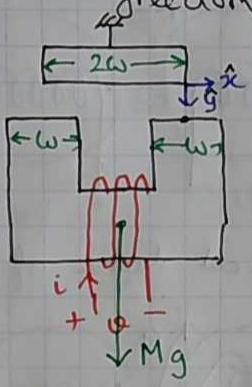
Zoom Password ?

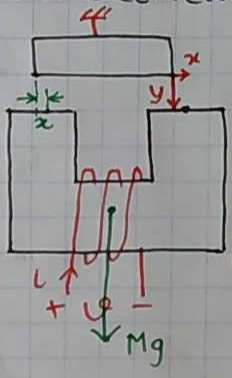
Last time: "Electrically" Non-linear systems.

→ graphical approach
ω_m, ω_m', fe.



→ Example of magnetic levitation.
- two mechanical degrees of freedom & one electrical port.





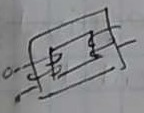
- Depth: D
- Fringing neglected
- $\mu_r \to \infty$

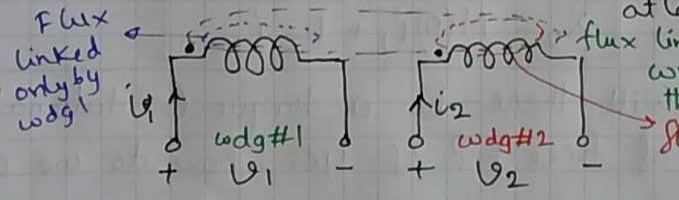
(Electrically linear).

* F_y^e will balance Mg
* F_x will restore lateral movement

Today: Q- How can we investigate multiple electrical ports and one mechanical port?

Let us go back to our transformer, except that now it is not on a magnetic core (for the time being at least).





L_1: Self inductance of wdg #1

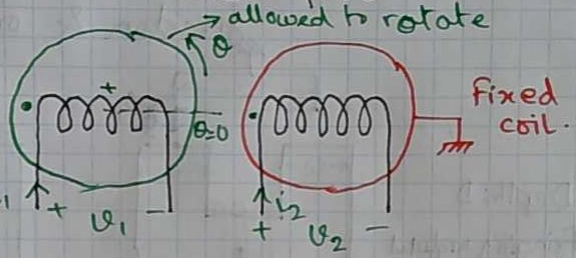
$$v_1 = \frac{d\lambda_1}{dt} \quad \text{and} \quad \lambda_1 = L_1 i_1 + M i_2$$

$$v_2 = \frac{d\lambda_2}{dt} \qquad \lambda_2 = M i_1 + L_2 i_2$$

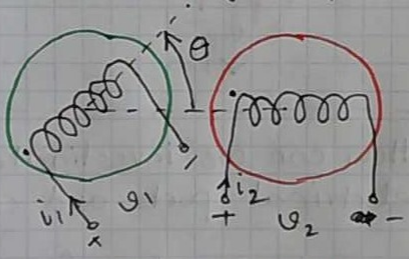
Q. What happens if we move, one of winding relative to another? What will change L_1, L_2, or M? or all?

– Mutual coupling will change, dependent on relative position of wdg 1 and 2.

So, let us add a mechanical port:



One mechanical degree of freedom, 'θ'.



Inherent assumption electrically linear system - & M are independent of current

$$\lambda_1 = L_1 i_1 + M(\theta)\, i_2$$

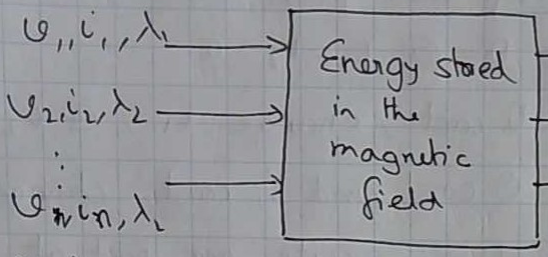
$$\lambda_2 = M(\theta) i_1 + L_2\, i_2$$

Q. Will there be a torque (rotational equivalent of force)? If yes, how do we calculate?

Parameters: L_1, M, L_2 Variables: $\lambda_1, i_1, \lambda_2, i_2$ and θ.

In general, N electrical port.

$\lambda_1, i_1, \lambda_2, i_2, \lambda_3, i_3 \cdots \lambda_n, i_n, \theta_1, \pi$



Multiple electrical ports

Multiple mechanical ports.

Conservation of energy process.

$$\frac{dW_m}{dt} = v_1 i_1 + v_2 i_2 + \cdots - f_1^e x_1 - f_2^e x_2 - \cdots - f_n^e x_2$$

$\frac{dW_m}{dt}$: Rate of change of stored energy.

For our example:

$$\frac{dW_m}{dt} = v_1 i_1 + v_2 i_2 - \tau_e \frac{d\theta}{dt}.$$

$$\Rightarrow dW_m = \int i_1 \, d\lambda_1 + \int i_2 \, d\lambda_2 - \int \tau_e \, d\theta.$$

Similarly, we can derive [See Sec. 4.6.3]

Coenergy:

$$dW_m' = \int \lambda_1 \, di_1 + \int \lambda_2 \, di_2 + \int \tau_e \, d\theta.$$

To compute torque, same basic steps.

- compute co-energy, W_m'
- compute torque, $\tau = \left.\frac{\partial W_m'}{\partial \theta}\right|_{\substack{i_1 = \text{const} \\ i_2 = \text{const}}}$.

Step 1: Computing co-energy.

Keep $\theta = \text{const}$.

$$W_m' = \int \lambda \, di \Rightarrow \int_0^{i_1} \lambda_1 \, di_1' + \int_0^{i_2} \lambda_2 \, di_2'$$

Choosing an integration path such that

first we change θ @ $i_1 = i_2 = 0$

next, we hold θ const., change i_1, @ $i_2 = 0$

next, we hold θ const and i_1 const, change i_2

bottomline: excite one part at a time.

θ = const. $i_2 = 0$ $\Big\}$ $W_m'_{(\theta, i_1, 0)} = \int_0^{i_1} \lambda_1 \, di_1 = \int_0^{i_1} L_1 i_1' \, di_1' = \frac{1}{2} L_1 i_1^2$

θ = const. i_1 = const i_2 - variable $\Big\}$ $W_m'_{(\theta, i_1, i_2)} = W_m'_{(\theta, i_1, 0)} + \int_0^{i_2} \lambda_2 \, di_2'$

$$= \frac{1}{2} L_1 i_1^2 + \int_0^{i_2} (M i_1 + L_2 i_2') \, di_2'$$

$$= \frac{1}{2} L_1 i_1^2 + M i_1 i_2 + \frac{1}{2} L_2 i_2^2$$

$$= \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M(\theta) \, i_1 i_2 .$$

$\therefore$ Electromagnetic torque.

$$\tau_e = \frac{\partial W_m'}{\partial \theta}\bigg|_{\substack{i_1 = \text{const} \\ i_2 = \text{const}}} = i_1 i_2 \frac{\partial}{\partial \theta} M(\theta)$$

Let us approximate $= M(\theta) = K \cos\theta$ (If we)

$\theta = 0^\circ$ max coupling

$\theta = 90^\circ$ zero. "orthogonal in space."

$$\therefore \boxed{\tau_e = -i_1 i_2 K \sin\theta}$$

$\theta = 0^\circ \Rightarrow$ no torque.

$\theta = 90^\circ \Rightarrow$ max torque acting as clockwise. ← synchronous.